



Efficient automated method to extract EOG artifact by combining Circular SSA with wavelet and unsupervised clustering from single channel EEG

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ABSTRACT

The emergence of affordable single-channel electroencephalography (EEG) headbands has introduced new possibilities for applications in brain-computer interface (BCI) systems and health monitoring. However, these EEG signals are often contaminated by electrooculogram (EOG) activity caused by eyelid blinking. This poses a challenge, especially in portable EEG devices that typically have limited or single EEG channels. The presence of eye-blink artifacts can mislead the diagnosis of brain state activity. Consequently, there is a growing demand to extract eye-blink artifacts from single-channel EEG signals. However, during the process of removing these artifacts from contaminated EEG signals, low-frequency artifact components in non-artifact regions are also inadvertently eliminated. In this work, an automated, robust and efficient framework is presented by combining Circular Singular Spectral Analysis (Ci_SSA) with wavelet analysis and unsupervised clustering. This framework aims to remove artifact segments while preserving the low-frequency content in non-artifact regions. Simulations were carried out on both synthetic and real data, and the results were compared with alternative methods. The performance of the proposed method is evaluated by metrics such as Relative Root Mean Square Error (RRMSE), Mean Absolute Error (MAE), Correlation Coefficient (CC) and Artifact Rejection Ratio (ARR) for synthetic signals, as well as Signal-to-Error Ratio (SER) and Artifact-to-Residue Ratio (ARer) for real EEG signals. After applying the proposed algorithm to filter the contaminated EEG signal, a noticeable enhancement in the detection accuracy of driver fatigue is observed as compared to the pre-filtering stage.

1. Introduction

The development of portable single-channel Electroencephalography (EEG) has revolutionized brain-computer interface (BCI) and health monitoring. The EEG signals are usually applied to interpret brain processes such as cognitive state and brain diseases like stroke and epilepsy. However, physiological artifacts like electromyogram (EMG), electrooculogram (EOG) and electrocardiogram (ECG) always contaminate the recorded EEG signals [1]. In long-term surveillance unlike other artifacts, EOG (eye-blink) artifacts are involuntary and so hard to avoid. As with any artifact, eye-blinks in EEG signals must be filtered before further processing to avoid wrong diagnosis of brain activity [2]. Generally, the low-frequency spectrum (0.5–12 Hz) of EEG signals often suffers from interference caused by the eye-blink artifact. In recent times, the growing prevalence of chronic diseases and the aging population has led to an increased demand for monitoring healthcare at home [3]. Several investigations have used in portable EEG equipment for various purposes, including investigation of Sleep issues and cognitive status in stroke patients, driver drowsiness applications [4,5]. Recently, portable EEG devices with fewer EEG channels,

such as single EEG channel equipment have been developed to reduce the strain and burden on the patient [6,7]. Artifacts in the multichannel EEG data have been removed using Blind source separation (BSS) techniques such as canonical correlation analysis (CCA) and independent component analysis (ICA) [8–12]. There introduced the hybrid methods, which can remove the artifacts in multichannel EEG. But these methods are not popular for single-channel EEG recordings to remove eye-blink artifacts. As a result, there is a need for new approaches that are adapted to processing the single-channel EEG signals.

An approach used to process single-channel EEG signals is an adaptive filter. The purpose is to extract eye-blink artifacts from EEG recordings, which was initially introduced in [13]. However, adaptive filter requires one reference signal to process the contaminated EEG signal. Hence, the contaminated signal is decomposed using the discrete wavelet transform (DWT). The low-frequency component obtained from this decomposition is then treated as the reference signal for the adaptive filter, enabling the removal of artifacts [14]. Along with these methods, data-driven decomposition algorithms like Empirical Mode Decomposition (EMD), Ensemble EMD (EEMD) combined

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with ICA and Detrended fluctuation analysis (DFA) are introduced to de-noise single channel EEG [15,16]. Although, this approaches provides inability to remove artifact region and impacts the non-artifact areas of the EEG signal. Recently, singular spectrum analysis (SSA) has been effectively used to decompose EEG signals. It is an efficient subspace-based technique for separating oscillating low frequency noise components from univariate signal [17]. The first eye-blink artifact removal using SSA has studied in [18]. The performance of SSA is enhanced with adaptive filter combined with new grouping techniques [19,20] and further it is used as smoothing filter [21]. In this method setting of threshold is key role for faithful separation of eye-blink artifacts. Recently new automation process was introduced to avoid threshold setting with new grouping based on Diffusion Entropy [22]. Here, Some content of the EOG artifact was lost due to mode mixing in decomposition stage and it also effects the information in non-artifact region of signal. Hence, in this paper, we present a novel framework that enhances the efficiency of automation processes. This is achieved by integrating Ci_SSA analysis, Morlet wavelet transform and unsupervised clustering. This framework effectively captures the eye-blink artifact while preserving the content of non-artifact regions in single channel EEG data. The performance metrics of the proposed method are assessed using both synthetic and real EEG signals, demonstrating significantly better results compared to existing methods. Furthermore, the effectiveness of the proposed method is further validated through its application in driver fatigue detection. The organization of paper is as follows: In Section 2. we summarized existing techniques such as, EMD, DWT, ICA, SSA based methods and present the working methodology of proposed method. In Section 3, we discuss about results with synthetic and real data set, and described driver fatigue detection application. Finally conclusion is addressed in Section 4.

2. Existed methods

2.1. EMD, DWT and ICA methods

Empirical mode decomposition (EMD) is an approach for analyzing non-linear and non-stationary time series in to different oscillations called intrinsic mode functions (IMFs). Noise related IMF component is identified by using threshold parameter. Recently, Detrended fluctuation analysis (DFA) is one of the suggested method to set a threshold based on scaling exponent [15]. In this process, any IMF component that exceeds a certain threshold is identified as an artifact and subsequently isolated. Another possible approach is Ensemble-EMD (EEMD) combined with ICA [16]. In this process generates the IMFs by considering number of ensembles as 50 and noise parameter as 1. Noise content IMFs are fed to *FastICA* method to generate Independent components (ICs) as $Z = UY$ with IMFs data matrix Y , mixing and unmixing matrices as A and U respectively. To extract the wanted signal, artifact content ICs are multiplied with respective column of matrix A gives eye-blink signal.

As discussed in [23], DWT-ICA can be implemented in two stages. In first stage, single channel EEG signal decompose in to several non overlapping spectral wavelet components by selecting proper order of wavelet. In second stage wavelet components are applied as input to ICA to generate artifact ICs. These ICs multiplied with mixing matrix gives wanted artifact signal.

2.2. SSA based methods

SSA based technique operating in four stages, namely, Embedding, Decomposition, Grouping and Diagonal averaging for artifact remove in single channel EEG [17]. Embedding converts the single channel data into multivariate data and apply singular value decomposition to get eigenvectors. These eigenvectors are clustered together during the grouping step and then undergo diagonal averaging, which involves

evaluating the signal of interest. Artifact component extraction is based on grouping process of eigenvectors based on mobility [21]. Another grouping technique discussed in Adaptive SSA (ASSA) based on adaptive grouping of Reconstruction components (RCs) by setting proper threshold [24]. A new automated grouping process has recently been introduced that utilizes Fluctuation based dispersion entropy [22]. The problem with this method is some of artifact content missed due to mode mixing and also influenced the non artifact region of EEG signal.

After examining traditional SSA methods, we have discovered a limitation in their effectiveness to accurately isolate the elementary series within EEG signals. Nevertheless, this drawback can be overcome by employing Ci_SSA. In Ci_SSA, the extraction of elementary components is accomplished through the grouping of frequencies. Conventional SSA methods face another limitation, which is the requirement for an optimal embedded dimension and appropriate threshold for the grouping process to extract EOG artifacts. However, in Ci_SSA, simply tuning the window length is sufficient to identify artifact signals using the first reconstruction component. This approach significantly improves the decomposition performance of EEG signals and effectively eliminates interference from EOG signals.

2.3. Clustering

All the existed algorithms extract the eye-blink artifact information at decomposition level and further it is attached to blind source separation methods to improve the eye-blink detection accuracy in raw EEG signal. However, as a result of the mode mixing effect, traditional algorithms were unable to properly segregate low-frequency EEG signal information, causing it to be present in non-artifact regions following decomposition. So that, Ci_SSA is the method used to estimate eye-blink information effectively in artifact region at decomposition stage and information in non artifact region preserves by isolating non-artifact information from eye-blink data by applying wavelet features based k-means unsupervised clustering. The most popular unsupervised machine learning approach for dividing a given data set into a collection of k groups is called “k-means clustering”, where k is the number of groups specified by user. The algorithm’s purpose is to locate cluster centers as widely apart as possible and to associate each data point with the nearest cluster center. In the k-means algorithm, the dissimilarity measure is commonly Euclidean distance. The procedure of clustering process is specified in Algorithm 1.

Algorithm 1 Basic K-means clustering algorithm

Input: X is a dataset having N points ,Desired number of clusters k .
Output: C is cluster of k set centroids.

procedure

Step 1: Initialize randomly k centroids.

Step 2: Each data point in X associate to nearest centroid. Hence data divided in to k clusters.

Step 3: Recalculate the centroid’s position.

Repeat Step 2 and Step 3 until cluster assignments stop changing.

End procedure

In this work clusters are extracted from k-means algorithm and it is used to separate artifact and non artifact regions.

2.4. Proposed methodology

The primary elements of the proposed approach for removing eye-blink artifacts are shown Fig. 1. It is a two-step procedure: first, using Circular SSA (Ci_SSA), an eye-blink artifact is identified from the contaminated single-channel EEG signal. Next step is denoise eye-blink artifact by using unsupervised clustering (k-means) algorithm based on time–frequency block of eye-blink data. Each stage is described in the sections that follow.

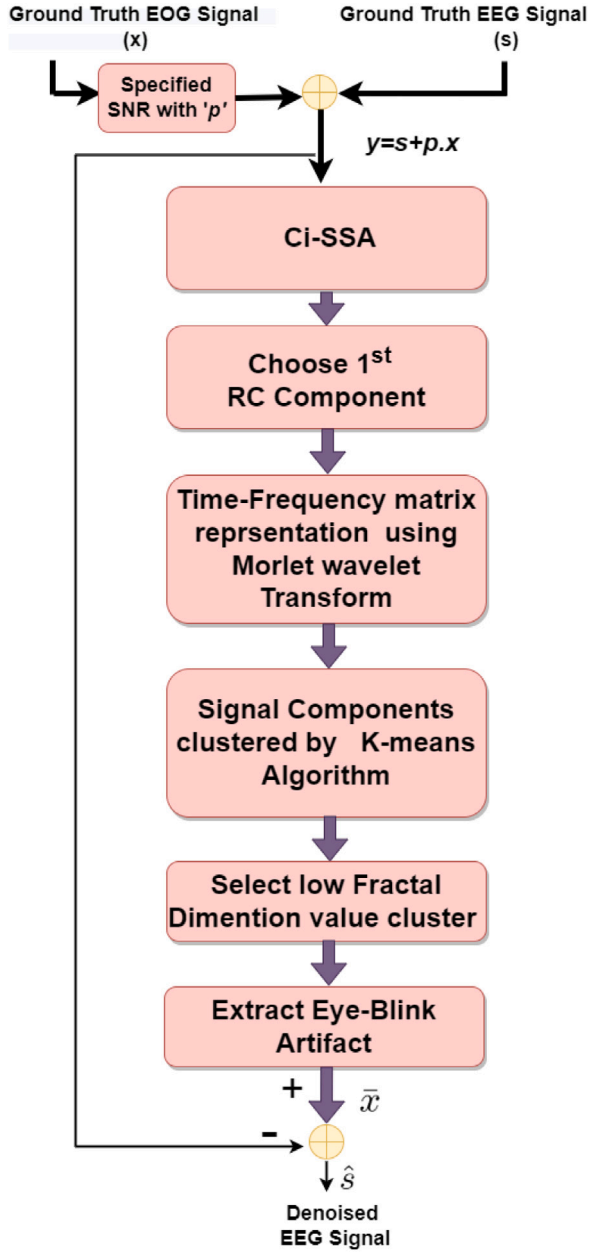


Fig. 1. Flow diagram of proposed method.

By employing Ci-SSA, the initial time series is partitioned into a set of predetermined frequency-based oscillatory components. This approach allows for precise identification of the extracted components, enabling users to arrange them based on their preferences. Let x_t be a zero-mean stationary process of length P and is given by $X = (x_1, \dots, x_P)'$. Here L be a positive integer and window length, so that $L < P/2$. Decomposition and reconstruction stages in SSA are repeated in this four-step algorithm.

Step 1- Embedding: We create a trajectory matrix by arranging usually lagged segments of size L from the original sequence. The $L \times R$ Trajectory matrix X , $R = P - L + 1$, given by:

$$X = \{x_1 | \dots | x_R\} = \begin{pmatrix} x_1 & x_2 & \dots & x_R \\ x_2 & x_3 & \dots & x_{R+1} \\ \vdots & \vdots & \dots & \vdots \\ x_L & x_{L+1} & \dots & x_P \end{pmatrix} \quad (1)$$

where $x_j = (x_j, \dots, x_{j+L-1})'$ is the vector of length L at time instance j .

Step 2- Decomposition: This stage will associate multiple frequencies with the trajectory matrix into elementary matrices of rank 1. To do so, we begin by constructing the circulant matrix, S_C , from the sample second moments as

$$\hat{\Psi}_m = \frac{1}{P-n} \sum_{i=1}^{P-n} x_n x_{i+n}, n = 0, \dots, L-1 \quad (2)$$

circulant matrix first row can be evaluated as

$$\hat{x}_n = \frac{L-n}{L} \hat{\Psi}_m + \frac{n}{L} \hat{\Psi}_{L-n}, n = 0, \dots, L-1$$

The eigenvalues can estimate from S_C as $\text{diag}(\lambda_1, \dots, \lambda_L) = \mathbf{V}^T S_C \mathbf{V}$, where $\mathbf{V}$ is Fourier unit matrix and $[\cdot]^T$ is complex transpose of $[\cdot]$. The eigenvector of q -column of $\mathbf{V}$, associated to eigenvalue λ_q is specified as $\mathbf{v}_q = L^{-1/2} (v_{q,1}, \dots, v_{q,L})$ with $v_{q,j} = \exp(-i2\pi(j-1)\frac{q-1}{L})$. Spectral density of the time series x_t is the estimated value of eigenvalue λ_q , for frequency $w_q = \frac{q-1}{L}, q = 1, \dots, L$. Hence, there is direct relation between eigenvalue λ_q and frequency w_q . i.e

$$\lambda_q \approx \hat{f}\left(\frac{q-1}{L}\right) = \sum_{n=0}^{L-1} \hat{\Psi}_n \exp\left(i2\pi n \frac{q-1}{L}\right) \quad (3)$$

Like wise the trajectory matrix X can be decomposed as sum of elementary matrices, that is,

$$X = \sum_{q=1}^L X_q = \sum_{q=1}^L \mathbf{u}_q \mathbf{w}_q' \quad (4)$$

where $\mathbf{w}_q = X' \mathbf{u}_q$

Step 3- Basic frequency grouping: In this step elementary matrices are grouped based on frequency. The power spectral density has symmetrical in nature and it leads to eigenvalues as $\lambda_q = \lambda_{L+2-q}$ and their eigenvectors are complex conjugate together. That is $\bar{\mathbf{u}}_q = \mathbf{u}_{L+2-q}$. Here, elementary matrices $X_q = X_{L+2-q}$ have same frequency and we grouped them together with respect to elementary pairs of frequency $G_q = \{q, L+2-q\}$ for $q = 2, \dots, M$, where $M = \frac{L+1}{2}$ with $G_1=1$ and $G_{\frac{L}{2}+1} = \frac{L}{2} + 1$, for L is even. Elementary matrices can compute with respect to frequencies as

$$\begin{aligned} X_{G_q} &= X_q + X_{L+2-q} \\ &= \mathbf{u}_q \mathbf{u}_q^* X + \mathbf{u}_{L+2-q} \mathbf{u}_{L+2-q}^* X \\ &= (\mathbf{u}_q \mathbf{u}_q^* + \bar{\mathbf{u}}_q \bar{\mathbf{u}}_q^*) X \\ &= 2(R_{\mathbf{u}_q} R_{\mathbf{u}_q}' + I_{\mathbf{u}_q} I_{\mathbf{u}_q}') \end{aligned} \quad (5)$$

where $R_{\mathbf{u}_q} = \text{real}(\mathbf{u}_q)$ and $I_{\mathbf{u}_q} = \text{imag}(\mathbf{u}_q)$. So both X_q and X_{G_q} are real values. Construct the Q disjoint groups I_j of the pairs B_q that have substantial or interesting frequencies w_q . Finally, the trajectory matrix recovered by the contribution of particular elementary frequency components G_q is represented as $X = X_{G_1} \dots X_{G_M}$.

Step 4-Reconstruction: Diagonal averaging [25] or Hankelization of matrix X_{I_j} is the final step to reconstruct series

$$\tilde{x}_t^j = H(X_{I_j}) \quad (6)$$

Reconstruction time series are generally called Reconstruction Components(RCs). These RCs contains the artifact and EEG subspace content. Eye-blink artifact RC has extracted and applied to wavelet for processing. Extracted eye-blink artifact signal contains some low frequency content in non-artifact region by filter using SSA approach. Eye-blink artifact is directly subtracted from contaminated EEG signals results gives loss of low frequency component in non artifact region. Therefore, removal of these oscillation's are required while subtracting eye-blink artifact from contaminated signal. By using new methodology, Low frequency data in non artifact region of eye-blink signal is extracted effectively.

In this process artifact RC component of SSA output signal is represented in time–frequency form using morlet wavelet transform and it produce result matrix as

$$\hat{B} = \begin{pmatrix} \hat{b}(1,1) & \hat{b}(1,2) & \dots & \hat{b}(1,M) \\ \hat{b}(2,1) & \hat{b}(2,2) & \dots & \hat{b}(2,M) \\ \vdots & \vdots & \dots & \vdots \\ \hat{b}(L,1) & \hat{b}(L,2) & \dots & \hat{b}(L,M) \end{pmatrix} \quad (7)$$

$$= (\tilde{b}_1, \tilde{b}_2 \dots \tilde{b}_N)$$

In this context, L represents the number of frequencies at which the wavelet is generated. Every column of matrix $\hat{B}$ represents the feature vector ($\tilde{b}$) of j th sample of $\tilde{b}$. Then, each column of matrix $\hat{B}$ is clustered and provides labels for every feature vector. Each label information specifies the particular feature vector of $\tilde{b}$. With this labels information, we cluster the signals into number C as follows

$$\tilde{b}^n(j) = \begin{cases} \tilde{b}(j) & \text{if } \tilde{b}_j \in C_n, n = 1, 2, \dots, N \\ 0 & \text{if } \tilde{b}_j \notin C_n \end{cases} \quad (8)$$

After clustered the signal $\tilde{b}$ as $(\tilde{b}^1, \tilde{b}^2 \dots \tilde{b}^C)$, eye-blink artifact associate component among these clusters are identified by using Fractal Dimension(FD) parameter [26]. Usually, for denoised artifact signal FD value is low. So eye-blink artifact associated component ($\tilde{x}$) is extracted based on FD value and it is subtracted from the contaminated signal (y) gives artifact free signal ($\hat{s}$).

3. Results

The performance of the proposed and existed methods are analyzed with contaminated data formed by combining synthetic eye-blink data and true EEG signal. In this work, we used a set of different parameters to evaluate the proposed method's performance on a synthetic and real EEG dataset. Synthetic contaminated signal y is formed by mixing of true EEG signals and synthetic eye-blink signal x with some proportion constant p .

(i) Signal to Noise Ratio(SNR)

Signal to Noise Ratio of measured signal y is represented as

$$SNR = \frac{RMS(s)}{RMS(p_x)} \quad (9)$$

(ii) Relative Root Measure Square Error (RRMSE)

Relative Root Mean Square Error between two signals s and $\hat{s}$ is defined as

$$RRMSE = \frac{RMS(s - \hat{s})}{RMS(s)} \quad (10)$$

Here s and $\hat{s}$ represents ground truth signal and estimated signal respectively. The parameter should be small for good estimation of signal.

(iii) Mean absolute error (MAE)

Mean absolute error (MAE) is another metric to measure the performance in frequency domain

$$MAE = \sum_{i=k}^l \frac{|P_s(i) - P_{\hat{s}}(i)|}{l - k} \quad (11)$$

where $P_s(i)$ is true and $P_{\hat{s}}(i)$ is corrected EEG power spectrum values. For good artifact removal, the estimated MAE value should be minimum.

(iv) Artifact Reduction Ratio(ARR)

This performance metrics quantifies the percentage of reduction of artifact from contaminated signal

$$ARR = \left(1 - \frac{\rho_{clean} - \rho_{after}}{\rho_{clean} - \rho_{before}}\right) \times 100 \quad (12)$$

where ρ_{before} is the correlation value between contaminated signal and true EEG signal, ρ_{after} is correlation value between estimated and true EEG signal. Here ρ_{before} value is set to 1 and for efficient artifact

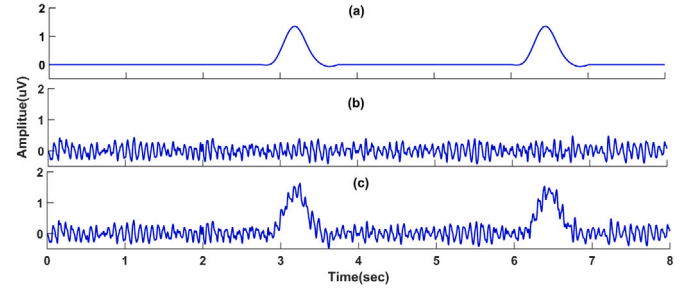


Fig. 2. (a). Synthetically generated EOG signal, (b). True EEG signal and (c). Contaminated signal.

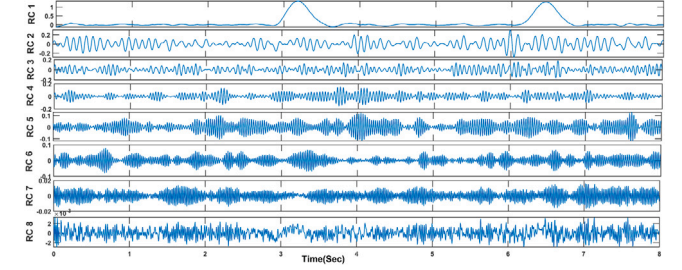


Fig. 3. Extracted RC components of contaminated signal of length 8 s by Ci_SSA.

removal ARR value should be high. The degree of associations between data variables is measured using correlation coefficients.

(v) Signal-to-Error Ratio(SER)

It quantifies EEG distortion outside of artifact region. Higher the value means better the performance and expressed in dB.

$$SER = 10 \log_{10} \left(\frac{E\{(y_c)^2\}}{E\{(d_c)^2\}} \right) \quad (13)$$

Where, $E\{[.]^2\}$ is estimation the variance of $[.]$, y_c and d_c are non-artifact segments of contaminated signal (y) and estimated eye-blink signals ($\tilde{x}$) respectively. As denominator d_c value low means higher is the SER, which gives more performance.

(vi) Artifact-to-Residue Ratio(AReR)

This parameter computed over artifact segments.

$$AReR = 10 \log_{10} \left(\frac{E\{(y_d)^2\}}{E\{(y_d - y_c)^2\}} \right) \quad (14)$$

Where, y_d and y_c are artifact region segments of contaminated signal (y) and estimated eye-blink signal ($\tilde{x}$) respectively. The approximated artifact y_c is nearly identical to the original y_d . i.e, the residue value ($y_d - y_c$) is as small as possible for good performance of AReR.

3.1. Simulation results with synthetic data set

In our analysis, we adopt the basic mixing model equation for a contaminated signal, which can be expressed as $y = s + p.x$. Here, the true EEG data s , sampled at a frequency of 250 Hz, is combined with a synthetically generated eye-blink artifact signal x through the mixing component p . The SNR value of the contaminated EEG signal is determined by the artifact mixing constant p . When the value of p changes, corresponding reverse changes can be observed in the SNR value. Fig. 2 shows the synthetically generated contaminated signal, which combines the true EEG data with an eye-blink artifact at a SNR of -4 dB. In Initial stage, eye-blink artifact is extracted by proper decomposition of contaminated signal using circular SSA and the first seven RCs represented in Fig. 3.

Here faithful reconstruction of eye-blink artifact depends upon proper arrangement of window length L . The window length represents the embedding dimension and it is essential parameter for the

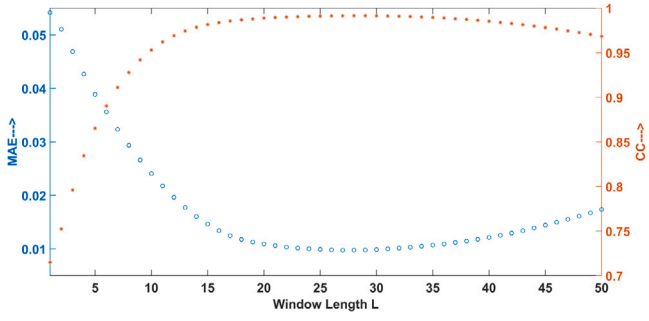


Fig. 4. Performance of proposed method in terms of CC and MAE for various window lengths L .

proposed Ci_SSA algorithm for proper decomposition to estimate eye-link component effectively. In the initial stage of the algorithm, MAE and CC of the signal are recursively computed at different values of L . To achieve better results, the L value was tuned to 28 for processing this specific dataset. By using this value, the denoised signal was able to produce a MAE value of 0.12 and a maximum CC value of 0.98. Graphically it is depicted in Fig. 4. In proposed method, initially Ci_SSA generated RC components with proper window length and choose first RC (RC1) as artifact signal, as it contains eye-blink information. This extracted artifact RC component contains additional low frequency information in non-artifact region of signal. In order to eliminate low frequency content in non artifact region, RC component passed over Morlet wavelet transform to generate Time-Frequency feature matrix of eye-blink artifact signal, which is then applied to k-means clustering algorithm. In the process, Wavelet transform computed in range 1–12 Hz with increment factor of 0.25. This computation produce a 45×2000 feature matrix information and it maps each sample of eye-blink artifact generated by Ci_SSA in to 45×1 size higher dimensional feature vector. Depending upon the requirement, clustering is formed from the information $(\bar{b}^1, \bar{b}^2, \dots, \bar{b}^C)$. Here for strong component reconstruction, we choose cluster number as 2. The eye-blink component in time-frequency feature metrics has strong information. As mentioned earlier, in the clustering process, the eye-blink and non-eye-blink feature metrics are divided into two clusters. That is, Eye-blink related information clustered as $\bar{b}^1$ and non eye-blink information clustered as $\bar{b}^2$. Usually fractal dimension value of eye-blink artifact to be low. So choose the cluster which have low FD value and it is considered as fully reconstructed eye-blink artifact component of single channel EEG signal.

Even though, EEMD+ICA, WICA, ASSA, FDisp based SSA methods extracted eye-blink information, they also contain low frequency information in non artifact region Fig. 5(a) and they offers poor reconstruction of EEG signal shown in Fig. 5(b). The frame work of FDisp based SSA was introduced in [22]. FDisp based SSA combined with wavelet features clustering with k-means method (FDisp based SSA+k-means) was computed in this paper and compare the results with proposed method to differentiate effectiveness of Ci_SSA over FDisp based SSA. While the recovery performance of the EEG signal using FDisp based SSA+kmeans may be comparable to the proposed method, it partially extracts the eye-blink artifact compared to the proposed method, as shown in the fourth and fifth rows of Fig. 5. It can also be observed from the power spectrum density (PSD) curves of the estimated and true EEG signals using both the proposed and existing methods. In comparison to other existing methods, the proposed method preserves the low-frequency content of the EEG signal. This can be visualized in the zoomed version of the PSD plots, as shown in Fig. 6. It is clearly observed that the curve of reconstruction signal using proposed algorithm is very much nearer to truth signal curve in (0.5–5) Hz band. The proposed approach applied on 19 synthetically contaminated data signals. The performance evaluation of proposed and existing

techniques over 19 samples record true EEG signals and plots for all simulated mean values of RRMSE, MAE, CC and ARR metrics are shown in Fig. 7. When the mixing component p is low, removal of eye-blink artifact is the problematic task. Here CC is computed with reference to synthetic Eye-blink artifact, whereas MAE, ARR and RRMSE are computed with true EEG signal. It is observed from Fig. 7.(a), (c), (d), that the proposed algorithm is effectively removed EOG artifact from contaminated signal and FDisp based SSA+kmeans gives comparative performance with proposed method in high SNR. In the case where the magnitude of the eye-blink artifact is high, particularly at low SNR levels, the performance of the proposed algorithm may be less effective. As with the parameter CC from Fig. 7(b), it is clear that the methods ASSA, FDisp based SSA, EMD-DFA and WICA are effectively estimate the blink artifact at low SNR and degrades performance in case of high SNR case. As from the observation, the EEMD+ICA method exhibits relatively lower performance compared to the existing methods

The aim of the automatic EEG artifact classification system is to correctly identify and remove artifacts in EEG signal datasets with minimal human intervention. Ideally, such models would facilitate EEG data processing without appealing to human experience and expertise in identifying artifacts or depending on other modes of artifact reduction. In this work Noisy EEG data decomposed by Ci_SSA with proper tuning window length (L) and processed with wavelet method combined with unsupervised clustering (k-means) to remove artifact without losing information in non-artifact region. Through out the denoising process, there is no parameter statically set by human. Hence, it improves the automation of the system.

3.2. Simulation results with real data set

The utilization of synthetic signal testing in this study serves the purpose of validating the artifact removal approach's effectiveness and facilitating a comparison with ground truth data. By employing this method, it becomes possible to quantify the algorithm's performance metrics, including SNR, MAE, RRMSE, and ARR, which assess its ability to eliminate EOG artifacts while preserving the underlying EEG signal. It also enables to optimize and fine-tune of method by iteratively adjusting parameters like window length L of Ci_SSA algorithm based on the performance results obtained from the simulated dataset. The efficacy of this fine-tuned method was evaluated using two databases. First EEG data were collected from a total of twenty-seven healthy subjects [27], comprising 14 males and 13 females. The electrode placement followed the 10–20 International System, with a total of nineteen EEG electrodes utilized. From these twenty-seven subjects, a total of fifty-four datasets were obtained, each with a duration of 20 s. The second database is called the EEG BCI Dataset [28], which was obtained from 55 participants who performed both rapid serial visual presentation (RSVP) and P300 speller tasks. This dataset contains real data and spans a duration of 285 s. The EEG data was collected using 32 Ag/AgCl active electrodes based on the international 10–20 system, which is a standardized method for electrode placement during EEG recordings. Within these datasets, the FP1 channel was identified as a contaminated signal caused by EOG and sample signal representation of two datasets are as shown in Fig. 8.

To facilitate the analysis, the long EEG signal was divided into multiple epochs, each with a duration of 20 s. These individual epochs were then subjected to the proposed method in order to enhance the effectiveness of the analysis. Fig. 9 displays the processed samples of 20 s duration epochs from two databases, using both the proposed and existing methods. We can see from Fig. 9(a), (b), unwanted low frequency components exist in non-artifact region are partially eliminated by filtering with methods EEMD+ICA, WICA, EMD-DFA, ASSA and Fdisp based SSA, whereas the filtered signal using FDisp based SSA+kmeans and proposed algorithms have matched content in non-artifact region of raw signal. In fact, no ground truth EEG signal is available to measure performance for real EEG signals. So, we manually identified

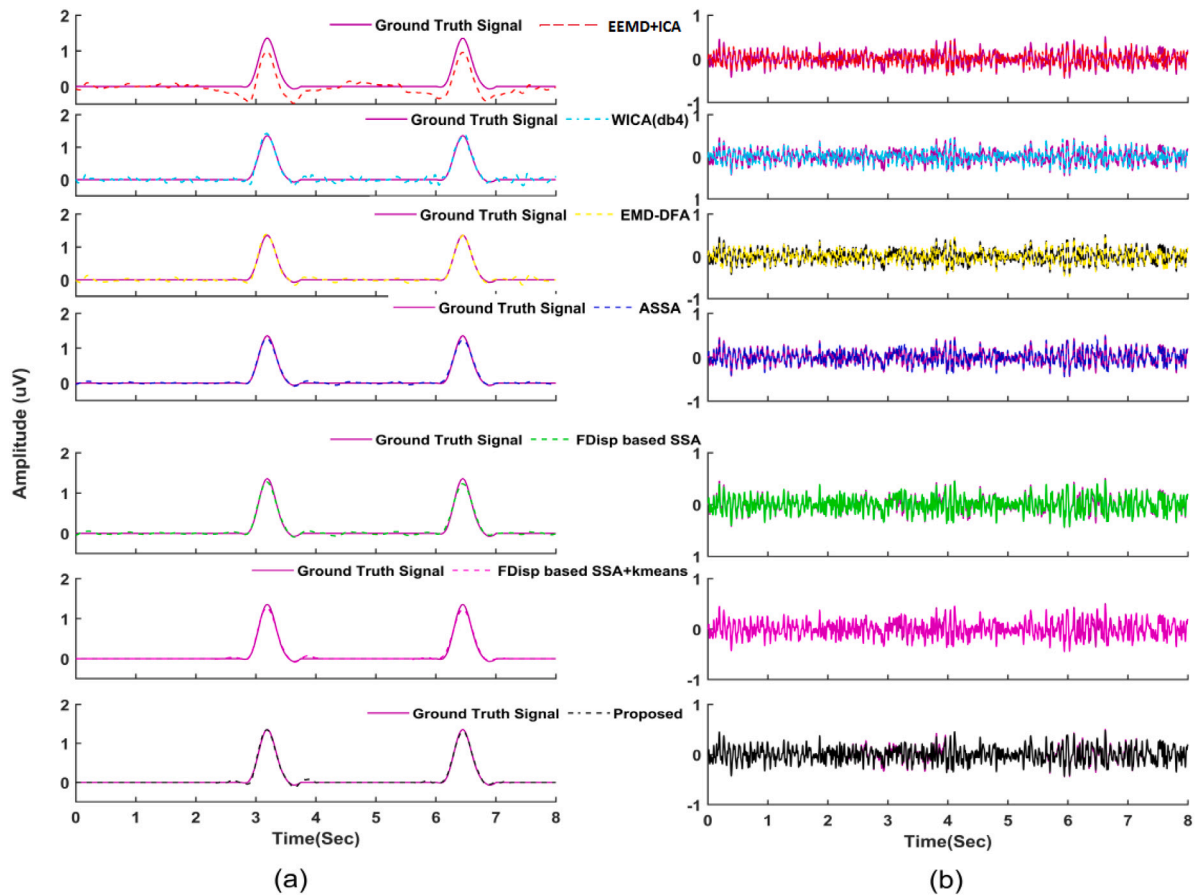


Fig. 5. Comparison plots of Estimated eye-blink artifact with ground truth EOG of existed and proposed methods.

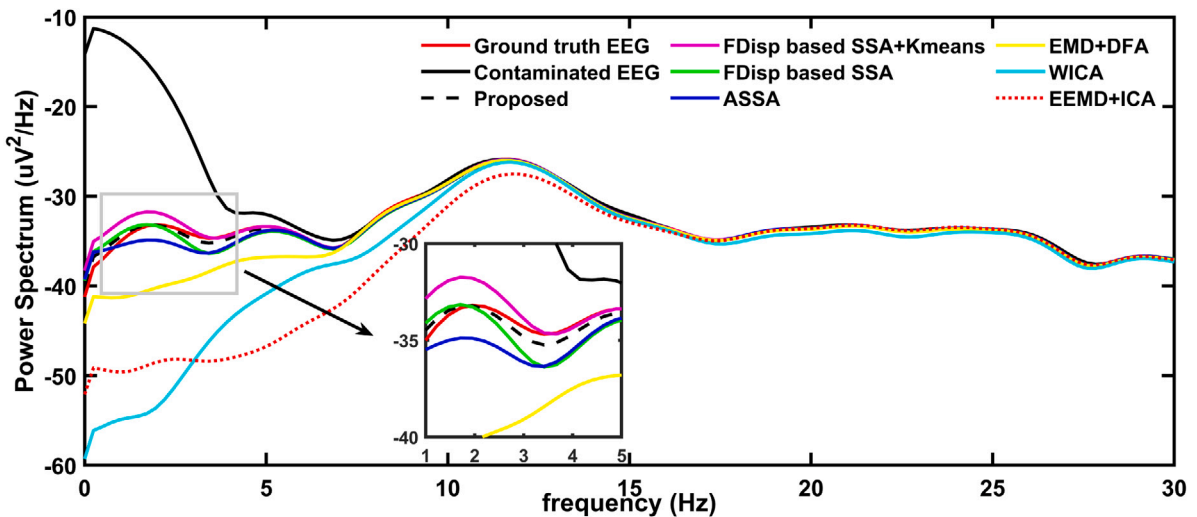


Fig. 6. Comparison Power spectrum representation of existed and proposed methods.

each record's artifact and non-artifact intervals and obtained SER and AReR values. Fig. 10 shows the improvement of artifact elimination with employing SER and AReR as complementary features besides the EEG after eye-blink removed by existed and proposed algorithms. As it shown from the computed mean values of quantified parameters for database-1 and database-2, the SER parameter of proposed algorithm (5.39, 10.30) and FDisp based SSA+kmeans (5.18, 8.08) methods produce phenomenal performance. It shows the information in non-artifact

region preserves effectively after denoise the contaminated signal compared to remaining methods FDisp based SSA (1.84, 3.27), ASSA (0.73, 2.26), EMD+DFA (0.61, 1.85), WICA (0.41, 2.50) and EEMD+ICA (0.75, 2.88). Another parameter AReR specifies ability of estimation of artifact in non-artifact region. Computed results represents proposed method (20.32, 12.47), ASSA (19.77, 11.79) and EMD+DFA (20.20, 10.81) are effectively estimated the artifact in contaminated region compared to FDisp based SSA+kmeans (13.29, 7.97), FDisp based SSA

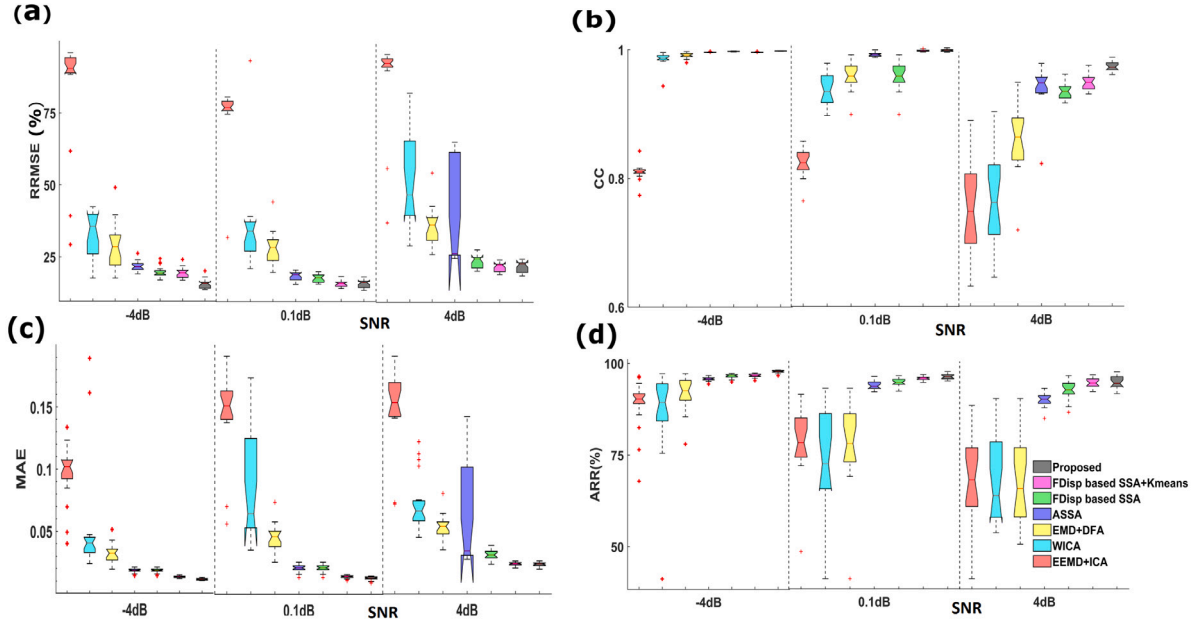


Fig. 7. Performance analysis of existed and proposed methods with metrics (a).RRMSE, (b).CC, (c).MAE and (d).ARR.

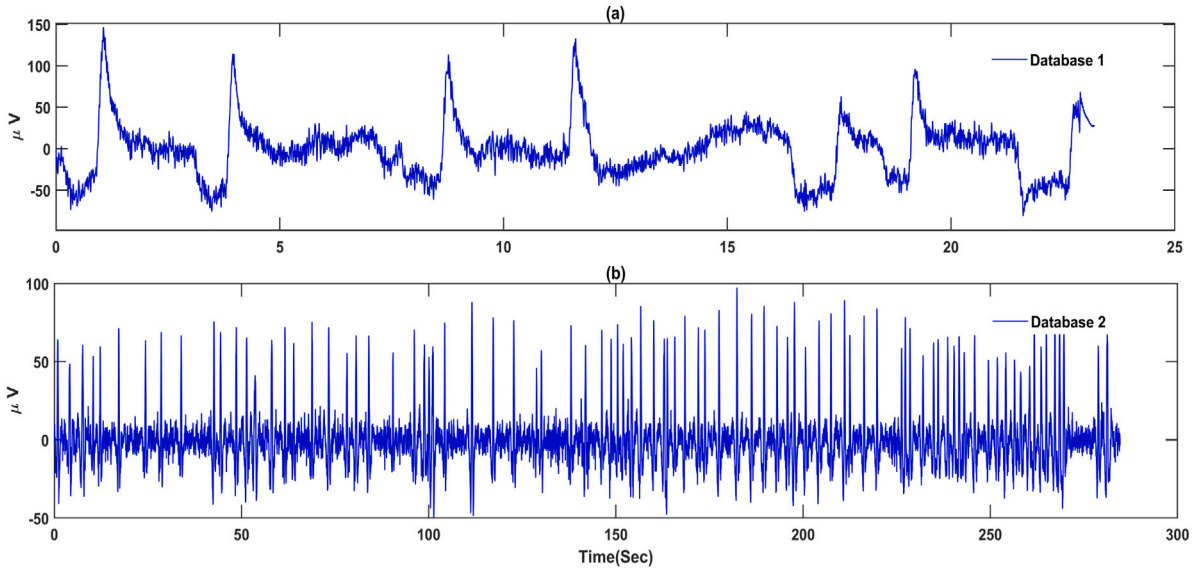


Fig. 8. Raw signal Samples of FP1 channel of (a). Mendely database(database-1) and (b).EEG BCI Dataset P300 Speller(database-2).

(12.78, 8.69), WICA (15.31, 10.71) and EMDD+ICA (17.62, 9.63). Finally from analysis, proposed method effectively estimate artifact component by preserving non-artifact component in EEG signal.

Power Spectral Density (PSD) evaluation is required in EEG analysis to examine the distribution of power across different frequency bands in the recorded brain signals. When it comes to the effect of EOG artifacts on EEG bands, the primary impact is on the low-frequency components, particularly in the delta and theta bands. EOG artifacts typically have higher power in these frequency ranges. Comparing the PSD measures before and after artifact removal helps in identifying the presence and magnitude of EOG artifacts in the recorded EEG data. If there is a significant difference in the PSD values after artifact removal, it indicates that the presence of quantity of EOG artifacts remains in denoised EEG signal. To analyze the PSD distribution in frequency bands of EEG signals, apply bandpass filters to the segmented EEG epochs to extract the desired frequency bands and measure PSD using Welch method to each filtered EEG epoch within the desired

frequency band [29]. Then ΔPSD measurements allow for a quantitative comparison between the PSD values before and after EOG artifact removal. To evaluate the degree of spectral distortion across the several frequency bands, delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), and beta (12–30 Hz) the following formula was used

$$\Delta PSD = |PSd_{in} - PSd_{out}| \quad (15)$$

Here PSd_{in} , PSd_{out} are power spectral densities before filtering and after filtering of contaminated EEG respectively [30,31]. A substantial decrease in power indicates successful removal of EOG artifacts and restoration of the underlying EEG signals. The computed mean ΔPSD of all subjects in database-1 and database-2 at all bands are computed and presented in Table 1 and Table 2 respectively. This evaluated metrics from Table 1, Table 2 can help to evaluate each algorithm's performance in both rejecting ocular artifacts and quantifying the induced distortion in the time domain. In database-1, the mean PSD of the contaminated signal is 23.66 (dB/Hz). However, after applying

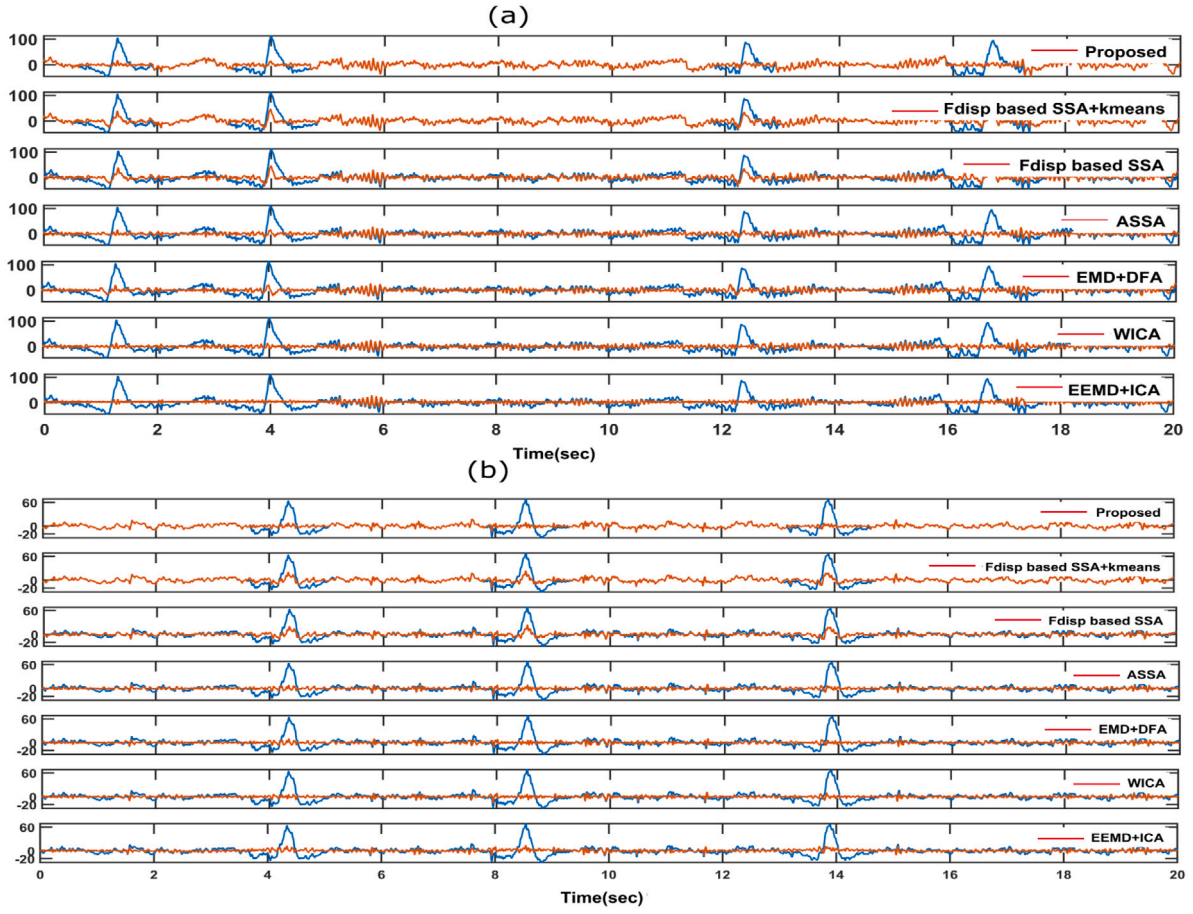


Fig. 9. Eye-blink Artifact removal with existed and proposed methods on 20 s sample from (a). Mendely database and (b).EEG BCI Dataset P300 Speller.

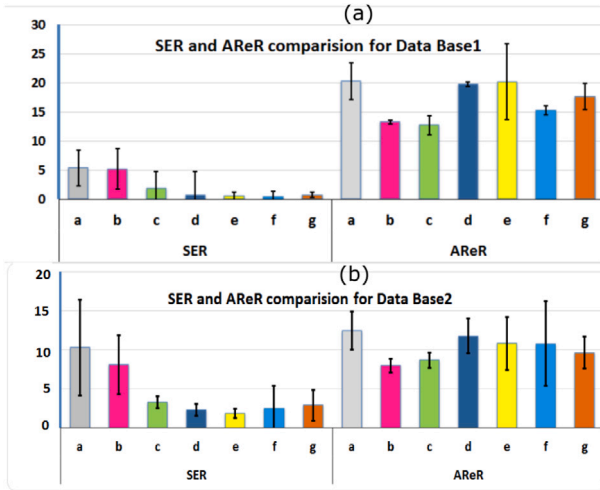


Fig. 10. SER and AReR performance metrics by the evaluation of Database-1 and Database-2 with proposed method (a), FDisp based SSA+kmeans(b), FDisp based SSA(c), ASSA(d), EMD+DFA(e), WICA(f) and EEMD+ICA(g) methods.

the proposed method to filter the signal, the estimated mean PSD of the denoised signals is 9.71 (dB/Hz). This indicates a significant difference of 13.95 (dB/Hz) compared to the PSD of the contaminated signal. Similar progressive results are obtained and specified in remaining bands as mentioned Table 1. As from the simulation results of database-2 quantified in Table 2, the proposed method attains less Δ PsD in Delta band (5.7 dB/Hz), Theta band (0.50 dB/Hz), Alpha band (0.20 dB/Hz)

and Beta band (0.010 dB/Hz), which shows the estimation of signal in all bands are much nearer to true signal. So that, proposed method successfully removes ocular artifacts while leaving the neural signals more unaltered in the time domain as compared to the other existed methods.

3.3. The driver fatigue detection

EOG artifact removal is a crucial step in driver fatigue detection systems that utilize EEG signals. These artifacts can contaminate EEG signals and introduce noise, making it challenging to accurately analyze the underlying brain activity associated with driver fatigue. EOG artifact removal is a significant role in driver fatigue detection by improving signal quality, enhancing feature extraction, increasing classification performance, and ensuring the reliability and real-world applicability of fatigue detection algorithms. By reducing the influence of EOG artifacts, these systems can more accurately assess and monitor driver fatigue, contributing to improved safety on the roads. To evaluate the performance improvement in driver fatigue (drowsiness) identification after eye-blink detection and removal with proposed method, EEG records were collected from [32]. These having 12 records of Fatigue state and Normal state data recorded in 40 channels each. Here 3rd channel is treated as EOG contaminated channel (FP1) of length 300760 and 301200 samples for Normal and Fatigue state respectively. Initially, the data is passed over band pass filter of (0.5–45) Hz and re-sampled to 250 Hz. Each channel is segmented into 60 epochs of 20 s long. Each epoch, alpha and beta band features namely PSD, Mean, Standard deviation, Sample entropy and Fuzzy Wavelet Packet Transform (WPT) are calculated for raw and filtered EEG by proposed method. The data formed 120×2501 features labeled

Table 1Quantitative metrics for Δ Psd of database-1.

Quantification metric Δ Psd	Proposed method	Fdisp based SSA + Kmeans	Fdisp based SSA [22]	ASSA[24]	EMD-DFA [23]	WICA [15]	EEMD-ICA [16]
Delta band (dB/Hz)	13.953 $\pm$ 1.24	14.746 $\pm$ 1.50	15.905 $\pm$ 1.45	16.481 $\pm$ 1.96	19.068 $\pm$ 1.21	21.871 $\pm$ 2.15	21.442 $\pm$ 2.10
Theta band (dB/Hz)	0.409 $\pm$ 0.21	1.151 $\pm$ 1.02	1.689 $\pm$ 1.10	1.715 $\pm$ 1.62	1.661 $\pm$ 1.20	2.976 $\pm$ 1.50	3.296 $\pm$ 1.39
Alpha band (dB/Hz)	0.073 $\pm$ 0.014	0.075 $\pm$ 0.021	0.113 $\pm$ 0.010	0.128 $\pm$ 0.24	0.157 $\pm$ 0.22	0.409 $\pm$ 0.31	0.576 $\pm$ 0.24
Beta band (dB/Hz)	0.017 $\pm$ 0.001	0.049 $\pm$ 0.002	0.126 $\pm$ 0.025	0.165 $\pm$ 0.002	0.085 $\pm$ 0.001	0.033 $\pm$ 0.002	0.134 $\pm$ 0.002

Table 2Quantitative metrics for Δ psd of database-2.

Quantification metric Δ Psd	Proposed method	Fdisp based SSA + Kmeans	Fdisp based SSA [22]	ASSA [24]	EMD-DFA [23]	WICA [15]	EEMD-ICA [16]
Delta band (dB/Hz)	5.71 $\pm$ 2.03	6.498 $\pm$ 2.15	22.684 $\pm$ 1.34	22.306 $\pm$ 2.11	20.805 $\pm$ 2.15	24.55 $\pm$ 2.14	26.165 $\pm$ 2.10
Theta band (dB/Hz)	0.505 $\pm$ 1.05	1.950 $\pm$ 1.54	4.259 $\pm$ 1.62	4.095 $\pm$ 1.65	4.086 $\pm$ 1.44	6.783 $\pm$ 1.54	6.718 $\pm$ 1.24
Alpha band (dB/Hz)	0.208 $\pm$ 0.707	0.224 $\pm$ 0.675	0.230 $\pm$ 0.70	1.004 $\pm$ 0.814	1.355 $\pm$ 0.784	1.305 $\pm$ 0.65	2.241 $\pm$ 0.545
Beta band (dB/Hz)	0.010 $\pm$ 0.0016	0.011 $\pm$ 0.0014	0.349 $\pm$ 0.027	0.361 $\pm$ 0.0026	0.373 $\pm$ 0.0025	0.587 $\pm$ 0.005	0.436 $\pm$ 0.004

Table 3

The values for classification metrics of Acc, Spe and Sen for Driver fatigue detection using proposed and FDisp based SSA+kmeans.

Rec. No.	Before filtering			Filtered by proposed algorithm			Filtered by FDisp based SSA + kmeans		
	Acc	Spe	Sen	Acc	Spe	Sen	Acc	Spe	Sen
1	68.20	66.67	65.25	75.83	75.00	76.67	68.33	65.34	67.32
2	94.17	96.67	91.67	97.50	98.33	96.67	88.40	90.25	92.40
3	71.67	56.67	86.67	75.20	61.80	88.70	72.50	65.60	86.60
4	81.67	76.67	86.67	84.20	81.20	86.00	79.60	78.56	79.50
5	72.50	71.67	73.33	77.50	80.00	75.00	77.50	80.00	75.00
6	82.50	76.67	88.33	85.83	76.67	95.00	83.70	74.60	74.40
7	70.00	66.67	73.33	70.83	75.00	66.67	68.33	65.34	67.32
8	72.10	58.33	81.67	80.00	70.00	90.00	79.17	70.00	88.33
9	52.50	46.67	58.33	65.00	63.33	66.67	64.17	61.67	66.67
10	69.17	65.00	73.33	69.17	51.67	86.67	68.33	51.67	85.00
11	7.50	6.67	8.33	65.00	63.33	66.67	64.17	61.67	66.67
12	67.50	65.00	70.00	82.50	86.67	78.33	70.83	75.00	74.00
Mean	67.46	62.78	71.41	77.38	73.58	81.09	73.75	69.97	76.93

Table 4

Comparison of proposed method with existing works on fatigue detection performance.

Method	Acc(%)	Spe(%)	Sen(%)
EEMD-ICA [16]	67.3	61.8	68.5
WICA [15]	67.8	63	73.6
EMD-DFA [23]	69.9	64.5	72.3
ASSA [24]	68.7	64.8	71.7
Fdisp based SSA [22]	70.6	64.5	72.8
FDisp based SSA+ kmeans	73.7	69.9	76.9
Proposed method	77.3	73.5	81

matrix and then it is applied to SVM classifier and used K-Fold Cross Validation as 10. To evaluate the performance of driver fatigue detection, specificity (Spe), sensitivity (Sen), accuracy (Acc) are calculated as follows

$$Sen = \frac{TP}{TP + FN} \times 100 \quad (16)$$

$$Spe = \frac{TN}{FP + TN} \times 100 \quad (17)$$

$$Acc = \frac{TN + TP}{FN + FP + TN + TP} \times 100 \quad (18)$$

. Where FN and TP are number of missed and correctly detected fatigue epochs, FP and TN represents false and correctly detected alert (Normal) epochs. In the comparative analysis of drowsiness detection between raw EEG and artifact-removed EEG signals, various performance metrics were evaluated, including accuracy, specificity, and sensitivity. The results presented in Table 3 indicates that the accuracy of drowsiness detection using artifact-removed EEG signals is 77.38%, while it is 67.46% with raw EEG signals. This signifies a

9.9% improvement in accuracy when utilizing artifact-removed signals. The specificity of drowsiness detection using artifact-removed EEG signals is 73.58%, whereas it is 62.78% with raw EEG signals. This suggests that the ability to correctly identify non-drowsy instances is 10.8% higher when artifact-removed signals are used as raw signal for diagnosis. The sensitivity of drowsiness detection using artifact-removed EEG signals is 81.09%, while it is 71.41% with raw EEG signals. This indicates a 9.7% increase in the ability to correctly identify drowsy instances when utilizing artifact-removed signals. So, as from fatigue detection scenario, filtered EEG signal metrics of proposed algorithm shows improvement in Acc (9.9%), Spe (10.8%) and Sen (9.7%) compared to fatigue detection metrics of before filtering. A comparable evaluation of Fdisp based SSA+kmeans with the proposed method are shown in Table 3. The fatigue identification accuracy of FDisp based SSA+kmeans are comparable with proposed method as Acc (73.7% vs 77.3%), Spe (69.9% vs 73.5%) and Sen (76.9% vs 81%). Accuracy of detection of fatigue existed evaluation methods Fdisp based SSA, ASSA and EMD+DFA quantifies 3.2%, 1.3% and 2.5% improvement respectively. There is not much considerable improvement observed by using WICA and EEMD-ICA methods as shown in Table 4. Efficiency of driver fatigue detection can improve by optimization of features selection and classifier parameters [33].

4. Conclusion

In this paper, new automated novel model has been introduced by combining Ci_SSA and morlet wavelet transform followed by unsupervised clustering to remove eye-blink artifact from single channel EEG by preserving the EEG information in non artifact region. To quantify performance of proposed model, simulations were done on one

synthetic and real and metrics were evaluated in terms of RRMSE, MAE, CC, ARR, SER and AeRR. Filtered signal with proposed method also improved the driver fatigue detection accuracy of 5.8%, as it compares the fatigue prediction performance before filtering. It can concluded that, this method effectively extract the eye-blink artifact in portable applications and improves the driver fatigue detection.

CRedit authorship contribution statement

Murali Krishna Y.: Methodology, Software, Validation, Investigation, Writing – original draft. **Vinay Kumar P.:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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